Robust Cross-Platform News Event Detection via Self-Supervised Modality Complementation

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***Abstract*—Multimodal news event detection aims to identify and categorize significant events across media platforms using multimodal data. Previous work was limited to a single platform and assumed complete multimodal data. In this paper, we explore a novel task of cross-platform multimodal news event detection to enhance model generalization for cross-platform scenarios. We propose a Self-Supervised Modality Complementation (SSMC) method to tackle the challenges of incomplete modalities and platform heterogeneity presented in this task. Specifically, a Missing Data Complementation (MDC) module is designed to overcome the limitations caused by incomplete modalities. It employs a separation mechanism that distinguishes between modality-specific and modality-shared features across all modalities, allowing for the augmentation of missing modalities with information extracted from common features. Meanwhile, a Multimodal Self-Learning (MSL) module addresses platform heterogeneity by extracting pseudo labels from the target platform's multimodal views and incorporating a self-penalization mechanism to reduce reliance on low-confidence labels. Additionally, we collect a comprehensive cross-platform news event detection (CNED) dataset encompassing 37,711 multimodal samples from Twitter, Flickr, and online news media, covering 40 public news events verified by Wikipedia. Extensive experiments on the CNED dataset demonstrate the superior performance of our proposed method.**

***Index Terms*—News event detection, cross-platform, multimodal data, social media.**

I. Introduction

In the realm of digital media and information dissemination, the rapid advancement and ubiquity of online platforms have transformed how news is reported, consumed, and shared. This digital revolution has led to the emergence of news event detection, a sophisticated task that involves analyzing data from various platforms, such as Twitter, Flickr, and online news media, to identify and categorize news events. News event detection plays a crucial role in enhancing the effectiveness of journalism [1], improving the efficiency of social media monitoring [2], and facilitating the management of public safety concerns [3]. For instance, in the context of disaster management, it can quickly aggregate reports from diverse sources, offering a comprehensive view of the event, which is vital for coordinating emergency responses.

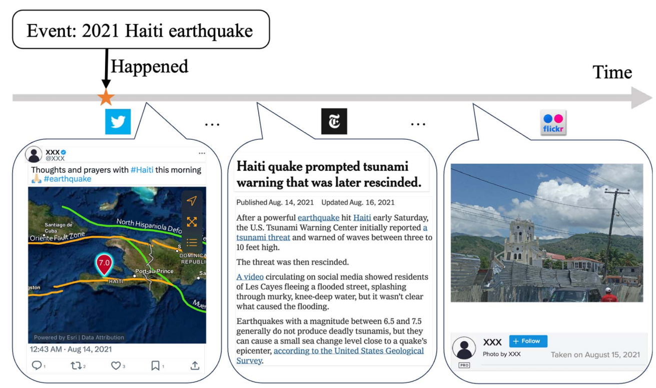

Fig. 1. As a news event develops, different platforms provide information from different perspectives for it.

Early efforts in news event detection [4], [5], [6], [7] predominantly relied on single-modal data, focusing either on text or images independently. For instance, Vavliakis et al. [6] defined the concept of important events and proposed an efficient methodology for event detection from large time-stamped web document streams. However, single-modal approaches often overlook the rich context provided by combining multiple data types. In contrast, multimodal data offers a more holistic view of news events, leading to enhanced accuracy and depth in detection. Recent research [8], [9], [10], [11] has thus shifted towards exploring multimodal news event detection. For example, Lin et al. [10] proposed a multi-modal fusion with external knowledge model that utilizes an attention mechanism to fuse external knowledge, images and text to detect news events.

Despite these advancements, the majority of existing research focuses on single-platform data, which limits the scope and diversity of the detected events. Real-world news events, however, manifest across multiple platforms, each offering unique perspectives and modalities of information. Consider the example illustrated in Fig. 1, where an event "2021 Haiti earthquake" emerges. The initial wave of information often comes from social media platforms like Twitter, offering immediate firsthand

Received 24 July 2024; revised 8 April 2025; accepted 18 July 2025. Date of publication 31 July 2025; date of current version 15 September 2025. This work was supported by Hong Kong Research Grants Council under the Collaborative Research Fund under Grant C1043-24G, in part by the Hong Kong ITC under the Innovative Technology Fund under Grant GHX/315/22GD, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515010237. Recommended for acceptance by R. Akbarinia. *(Corresponding author: Zhenguo Yang.)*

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The dataset is available at https://github.com/RetrainIt/CNED.

Digital Object Identifier 10.1109/TKDE.2025.3594200

accounts and public reactions. This is soon followed by the more structured and analytical coverage provided by online news media platforms, such as The New York Times, which brought in-depth reports and analyses. In parallel, other platforms like Flickr capture and share visual narratives through photographs. This progression underscores the necessity of cross-platform detection to gain a comprehensive and nuanced understanding of news events. However, current research methods tend to falter when confronted with cross-platform scenarios, demonstrating diminished effectiveness and adaptability in these more complex environments.

In this paper, we delve into the novel task of cross-platform multimodal news event detection, aiming to enhance the precision of news event detection across different platforms. This task seeks to leverage data from source platforms with known event labels alongside unlabeled data from target platforms, training models to perform more effectively on unlabeled target platform data. However, it introduces several challenges, i.e., incomplete modalities, platform heterogeneity and the scarcity of annotated datasets. Therefore, we propose a Self-Supervised Modality Complementation (SSMC) approach designed to tackle these challenges. Specifically, our method is underpinned by two components: a Missing Data Complementation (MDC) module and a Multimodal Self-Learning (MSL) module. The MDC module employs a modality classifier to differentiate between modality-specific and modality-shared features across all modalities. When a modality is absent, it becomes possible to supplement it using the information contained within the common features of another modality. The MSL module addresses platform heterogeneity by leveraging self-learning (i.e., pseudo-labeling), which individually exploiting the relationships of semantic, image, text, and joint multimodal features in different spaces. In particular, it uses a nearest-neighbor clustering algorithm to achieve multi-views pseudo labeling and utilizes high-confidence pseudo labels for self-learning while self-penalizing low-confidence pseudo labels. In addition, we collect a cross-platform news event detection (CNED) dataset for the cross-platform multimodal news event detection task. Specifically, it contains 37,711 multimodal samples covering 40 public news events from three distinct platforms, i.e., Twitter, Flickr and online news media. Each event within our dataset is verified through Wikipedia, ensuring reliability and breadth of coverage across a wide range of topics. This dataset not only facilitates the exploration of cross-platform multimodal news event detection but also sets a new benchmark for future research in this domain. The experimental results on the CNED dataset demonstrate the effectiveness of our proposed SSMC method.

Our contributions are summarized as follows:

- We propose a Self-Supervised Modality Complementation (SSMC) method that effectively addresses the challenges of incomplete modalities and platform heterogeneity in cross-platform multimodal news event detection.
- We compile a comprehensive Cross-platform News Event Detection (CNED) dataset, bridging the gap in multimodal data resources across diverse platforms.
- Through extensive experiments on the CNED dataset, we validate the effectiveness of our SSMC method, setting a new benchmark for cross-platform multimodal news event detection.

The rest of the paper is organized as follows. Section II reviews the related works. Section III introduces the preliminaries and problem statement. Section IV presents the details of the proposed SSMC approach. The experiments are conducted in Section V, and the conclusion is presented in Section VI.

II. Related Work

In this section, we briefly review the related work on news event detection, missing modality problem and domain adaptation.

A. News Event Detection

The domain of news event detection has significantly evolved from its origins in topic detection and tracking [12]. Initial investigations predominantly concentrated on single-modal event detection [13], [14], [15], [16], laying the foundational methodologies. For instance, Guo et al. [13] innovated a hierarchical neural framework tailored for event recognition within personal photo collections, marking a milestone in the field. Hu et al. [15] employed a neural model grounded in long short-term memory to facilitate the learning of shared event representations across various tasks. Progressing beyond, the focus shifted towards multimodal news event detection [8], [9], [17], [18], [19], [20], incorporating diverse data types to improve detection accuracy. Notably, Yang et al. [9] harnessed video and title metadata from YouTube, demonstrating the potential of combining modalities. Moreover, Qian et al. [8] introduced an open-world social event classifier model, leveraging both image and textual content, highlighting the complexity of capturing events in an open-world setting. Recently, Li et al. [17] attempted to mitigate this limitation by proposing an adaptive transformer-based conditioned variational autoencoder network, aiming to adaptively manage the missing modality issue. Nonetheless, their solution primarily caters to single-platform applications, revealing a gap in cross-platform detection.

B. Missing Modality Problem

The missing modality problem poses significant challenges in the field of cross-platform multimodal news event detection and other multimodal applications. Research in this area has predominantly followed two approaches to mitigate the impact of missing modalities: 1) Generative methods [17], [21], [22], [23] focuses on using generative models to predict or recreate the missing modalities based on the modalities that are available. These methods leverage the power of generative models to fill in the gaps where data is incomplete, thereby ensuring that the system can still perform its intended function despite the absence of some modalities. For instance, Suo et al. [22] introduced a new framework focused on learning patient similarity from multi-modal healthcare data, even when some of those modalities are missing. Li et al. [17] utilized a conditioned variational autoencoder for generating the missing modalities, thereby facilitating event detection. 2) Joint multimodal representation learning [24], [25], [26] focuses on creating a unified representation that contains information from all available modalities, even in the absence of some. This kind of method aims to leverage the shared information across modalities to infer missing data and maintain performance. For example, Ma et al. [25] proposed using Bayesian meta-learning to estimate the latent features of data. Similarly, Lee et al. [26] explored multimodal prompting with missing modalities for visual recognition. However, these methods were primarily developed for scenarios with three or more modalities, where the remaining modalities can provide complementary information when one is missing. This becomes particularly challenging in bimodal scenarios such as cross-platform multimodal news event detection, where the absence of one modality leaves only a single information source without redundancy, creating a more difficult problem than in trimodal settings [27], [28].

C. Domain Adaptation

Domain adaptation techniques aim to bridge the gap between the source domain (the domain for which the model has labeled data) and the target domain (the domain for which the model needs to make predictions but has limited or no labeled data). By leveraging these techniques, it is possible to enhance the ability of a cross-platform multimodal news event detection model to generalize across platforms and modalities, thereby effectively mitigating the issues posed by platform heterogeneity. Existing domain adaptation methods can be categorized into three classes. The first class employs distance metrics to reduce modality discrepancies [29], [30], [31], e.g., deep adaptation network [30] used maximum mean discrepancy to minimize the distance between source and target domains. The second class leverages adversarial learning to learn modality invariant features [32], [33], [34], e.g., domain-adversarial neural networks [34] confused the modality discriminator to learn modality invariant features. The third class utilizes pseudo labeling to achieve self-training in the target domain [35], [36], [37], [38], e.g., auxiliary target domain-oriented classifier [36] utilized prototypes to obtain pseudo labels for self-training. Despite the effectiveness of these approaches in bridging domain discrepancies, especially in image recognition tasks, they exhibit limitations when confronted with significant domain gaps (especially in multimodal scenarios), often leading to negative transfer. Moreover, the challenge of missing modalities remains inadequately addressed.

III. Preliminaries and Problem Statement

A news event is a significant occurrence or happening that is the subject of media coverage and public interest. In the digital era, such events are disseminated across various platforms, each with its unique presentation and content format. Cross-platform multimodal news event detection is the process of identifying and categorizing these news events across different platforms by analyzing multimodal data, such as text articles and images. A more formal definition of the problem is illustrated as follows.

Problem 1 (Cross-platform Multimodal News Event Detection): Cross-platform multimodal news event detection aims to address the identification of news events y^T in a target platform's dataset $\mathcal{D}^T$ based on the learning from a source platform's labeled dataset $\mathcal{D}^S$ and the target platform's unlabeled dataset $\mathcal{D}^T$. Formally, the source dataset is denoted as $\mathcal{D}^S = \{(I^S, T^S, Y^S)\} = \{(i_j^S, t_j^S, y_j^S)\}_{j=1}^{n_S}$, consisting of n_S samples, where $i_j^S \in \mathbb{R}^{d_{SI}}$ and $t_j^S \in \mathbb{R}^{d_{ST}}$ are the image and text modalities of the j-th sample, respectively, and $y_j^S \in \mathcal{Y} = \{1, 2 \ldots, C\}$ is the labeled news event. The target dataset, lacking such labels, is represented as $\mathcal{D}^T = \{(I^T, T^T)\} = \{(i_j^T, t_j^T)\}_{j=1}^{n_T}$ with n_T samples. Notably, in some samples, either the image I or the text T from $\mathcal{D}^S$ and $\mathcal{D}^T$ might be missing. The goal is to predict the event labels $y^T \in \mathcal{Y}$ for $\mathcal{D}^T$, effectively bridging the gap between the multimodal data representations across platforms.

TABLE I
The Statistics of CNED Dataset

	#Event	#Sample	#Words	#Language
Twitter (T)	40	24,607	14.72	109
Flickr (F)	40	9,191	46.66	51
Online News (O)	40	3,913	756.95	26

IV. Methodology

A. Overview of the Framework

As illustrated in Fig. 2, our SSMC model first uses CLIP [39] to extract the features of images and text separately. For each modality data, we design a modality-specific layer, i.e., F_T for text and F_I for image, to extract the specific modality information, and a modality-shared layer, i.e., F_C, to extract the common information across different modalities. Then, we concatenate them with a residual module to fuse these two parts of features. Finally, we use a fusion module to combine two modal features and then proceed to classification. When a modality is missing, e.g., text in the target domain, the text modality-specific layer F_T becomes unavailable. We supplement it with the common features extracted from image modality, which not only plays a complementary role but also ensures end-to-end training. For the target platform, we utilize the semantic pseudo labels obtained from the output of the model and structural pseudo labels extracted from the multimodal common features (e.g., E_{ic}^T) and the multimodal fusion features (i.e., E^T) to perform high-quality pseudo label screening to achieve self-learning.

B. Data Preprocessing

Different news events can originate from various countries, thereby presenting a multilingual challenge in relevant social media. As shown in Table I, there are 109 languages in the posts on Twitter about 40 different news events. We directly use Google Translation API[1] to convert multiple languages into English as our approach focuses on incomplete modalities and platform heterogeneity. Furthermore, to bring the multimodal feature distributions closer, we utilize CLIP [39] with ViT-B-32 for image and text feature extraction. The extracted features can be represented as E_i^S, E_i^T, E_t^S and E_t^T.

[1]https://cloud.google.com/translate

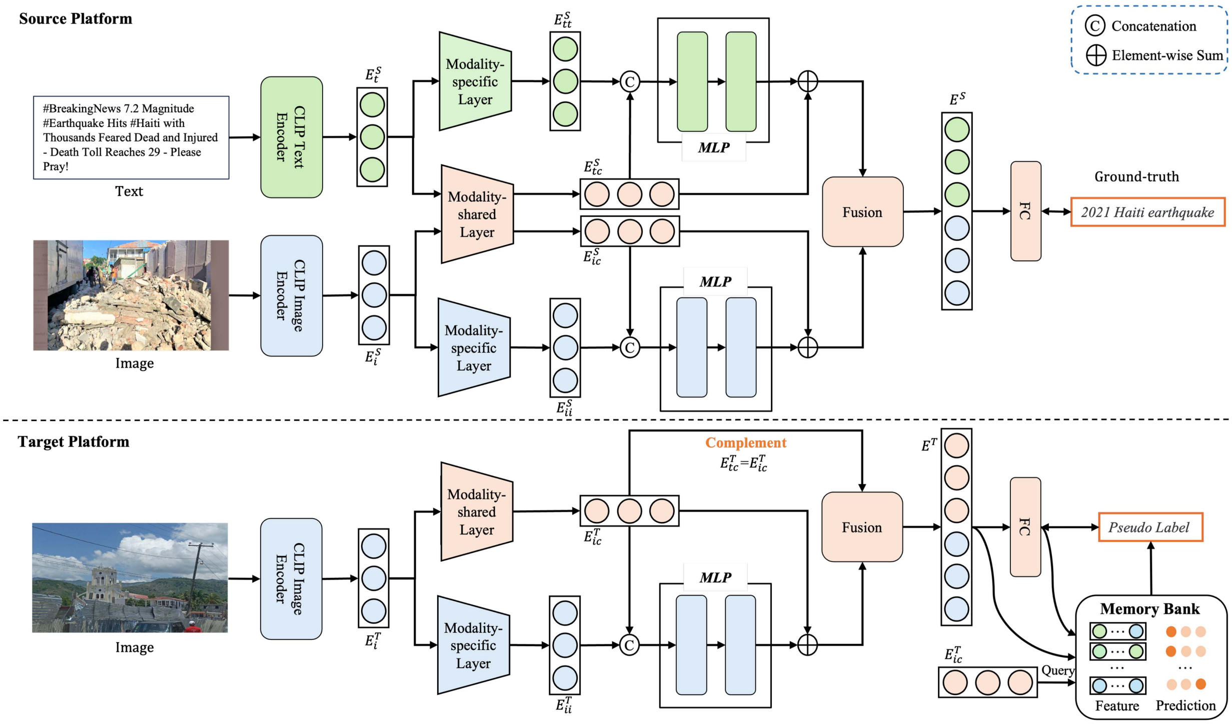


Fig. 2. An overview of the proposed SSMC method when the text of the target domain is missing. The upper flow represents the source platform with labeled data (exemplified by Twitter posts with both images and accompanying text); and the lower flow depicts the target platform with unlabeled data (Flickr for example, where images are prevalent without text). Best viewed in color.

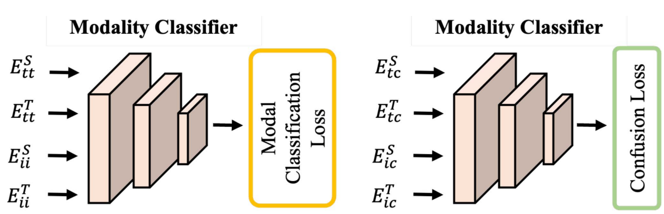


Fig. 3. Missing Data Complementation Module.

C. Missing Data Complementation (MDC)

Missing modality is common in multimodal learning. We consider supplementing the missing modality with information from another modality. However, there exists a modality gap between different modalities of data. To mitigate this gap, we design a separation mechanism to obtain modality-specific and modality-shared features between modalities, and then use the common features to supplement information when a modality is missing. Specifically, we utilize two modality-specific layers to extract text-specific and image-specific features respectively, and one modality-shared layer to extract information common to both modalities.

To achieve this, we design two losses, i.e., modal classification loss and confusion loss, as shown in Fig. 3. Specifically, the modal classification loss assumes that if different modalities of data can be classified as that modality, then the modality contains modality-specific features, which can be expressed as:

$$\ell_{mc} = -\sum_{j=1}^{|\mathcal{D}_S+\mathcal{D}_T|} \sum_{m\in\{I,T\}} \left(d^{(m)}\right)^\top \log\left(F_{\mathrm{mc}}\left(E_m^{(j)}\right)\right), \quad (1)$$

where $d^{(m)}$ represents a [1,0] or [0,1] vector when the input modalities are images (I) or text (T), respectively. F_{mc} denotes the modality classifier, which is composed of a multilayer neural network. $E_m^{(j)}$ refers to the j-th features extracted by modality-specific layers, i.e., E_{ii}^S, E_{ii}^T, E_{tt}^S and E_{tt}^T.

The confusion loss is used to achieve the goal of common feature extraction by confusing the modality classifier F_{mc}, which can be represented as:

$$\ell_d = -\sum_{j=1}^{|\mathcal{D}_S+\mathcal{D}_T|} \sum_{m\in\{I,T\}} \left(u^{(m)}\right)^\top \log\left(F_{\mathrm{mc}}\left(E_m^{(j)}\right)\right), \quad (2)$$

where $u^{(m)}$ represents a [0.5, 0.5] vector. $E_m^{(j)}$ refers to the j-th features extracted by the modality-shared layer, i.e., E_{ic}^S, E_{ic}^T, E_{tc}^S and E_{tc}^T.

The adversarial relationship between these two losses, ℓ_{mc} and ℓ_d, is essential for preventing information leakage between modality-specific and modality-shared features. While ℓ_{mc} encourages the modality-specific features to be distinguishable by maximizing the log probability of correct modality classification, ℓ_d simultaneously pushes the shared features to be modality-agnostic by enforcing a uniform probability distribution [0.5, 0.5] across modalities. This creates an effective information bottleneck: any modality-distinguishing information that leaks into shared features would make them discriminable by the modality classifier F_{mc}, thereby increasing ℓ_d. Through this adversarial mechanism, modality-specific information is constrained to remain in the respective modality-specific encoders.

When the modality is complete, we concatenate the modality-specific features and common features for each modality, use a multilayer perceptron (MLP) for dimension reduction, and add the common features as a residual to form semantically complete modality embeddings for the text domain E_{text} and image domain E_{image}, which can be represented as:

$$E_{text} = \text{MLP}(\text{Concat}(E_{tt}, E_{tc})) + E_{tc}, \quad (3)$$

$$E_{image} = \text{MLP}(\text{Concat}(E_{ii}, E_{ic})) + E_{ic}, \quad (4)$$

where $\text{Concat}(\cdot,\cdot)$ represents the concatenation operation. For data with missing modalities, we compensate by extracting common features from another modality, e.g., $E'_{text} = E_{ic}$ if the text is missing. Finally, we obtain multi-modal features (i.e., E^S and E^T) by fusing E_{text} and E_{image} for different platforms, and then use a shared classifier F_S for classification. Specifically, concatenation is utilized as the fusion method.

D. Multimodal Self-Learning (MSL)

A significant domain gap exists across different platforms, especially in multimodal data. We consider using the self-learning method (i.e., pseudo labeling) because it is more effective when facing a large domain gap. The MSL module is designed to achieve adaptive modality through three key components: 1) multi-view pseudo-labeling that generates complementary semantic and structural labels from different feature spaces, 2) a multi-view consensus mechanism that ensures reliable supervision by checking consistency across available modalities, and 3) a self-penalization mechanism that leverages partial information from inconsistent samples. This design allows our model to adapt to scenarios where certain modalities may be missing during inference.

Our multi-view pseudo-labeling component generates two types of complementary pseudo-labels. First, we extract semantic pseudo-labels directly from the classifier's predictions:

$$\hat{y}_j^{se} = \arg\max_k p_{j,k}, j = 1, 2, \ldots, n_t, \quad (5)$$

where $p_j = F_S(E_j^T)$ is the C-dimensional prediction, F_S denotes the shared classifier in the last layer, and $p_{j,k}$ represents the k-th element of vector p_j, corresponding to the predicted probability or score for the j-th sample belonging to class k.

However, the reliability of semantic pseudo labels is limited due to the domain gap. To address this, we extract structural pseudo labels based on the manifold assumption theory [40], which posits that samples with similar semantic meanings cluster together in feature space, even across different domains [36]. Our approach generates structural pseudo-labels from three complementary feature spaces—image, text, and multimodal representations. This multi-view strategy serves two critical purposes: 1) it captures different perspectives of the same data, offering richer supervisory signals than single-view approaches, and 2) it ensures robustness when certain modalities are missing during inference, as we can still obtain reliable structural labels from any available modality.

To generate structural pseudo-labels, we first establish a memory bank to update the target domain features (including image common features E_{ic}^T, text common features E_{tc}^T and multimodal features E^T) and output probabilities of the model p_j. Specifically, we sharpen each probability and feature of the output via temperature scaling (i.e., $p_{j,k}^d = p_{j,k}^{\frac{1}{t}} / \sum_k p_{j,k}^{\frac{1}{t}}$ where t is the temperature scaling parameter of sharpening) and L2-normalization, respectively. During the training, we utilize these generated features and probabilities to update the memory bank by the moving average strategy. When the missing modality occurs, the memory bank only updates the features of the available modalities.

As shown in Fig. 4, we utilize the features from the target domain to retrieve the memory bank for each training step. Specifically, we use cosine distance to obtain the k-nearest samples for each sample from different feature spaces, i.e., image space E_{ic}^T, text space E_{tc}^T and multimodal space E^T. Furthermore, we find out the probabilities of the nearest neighbor samples and get their corresponding probabilities in the memory bank. Finally, we average the probabilities of the K nearest neighbor samples to obtain the final output probabilities for each

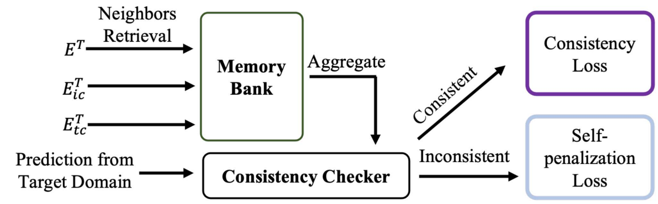


Fig. 4. Multimodal Self-learning Module.

Algorithm 1: SSMC Algorithm.

Input: Labeled source platform:
$\mathcal{D}^S = \{i^S, t^S, y^S | i^S \in I^S, t^S \in T^S, y^S \in Y^S\}$,
unlabeled target platform:
$\mathcal{D}^T = \{i^T, t^T | i^T \in I^T, t^T \in T^T\}$, the feature extractors F_T, F_I and F_C, the modality classifier F_{mc} and the shared classifier F_S.
Output: News event Y^T from target platform.
while $t \leq$ **MaxIter**
1: Compute the features E_i^S, E_i^T, E_t^S and E_t^T by the CLIP model.
2: Compute the multi-modal features E^S and E^T according to (3) and (4).
3: Compute the pseudo-labels for the target domain from different views according to (5) and (7).
4: Perform a consistency check for these pseudo-labels and compute the consistency loss ℓ_{co} and self-penalization loss ℓ_{sp} according to (8) and (9), respectively.
5: Compute the cross-entropy loss ℓ_c, modal classification loss ℓ_{mc} and confusion loss ℓ_d according to (11), (1) and (2), respectively.
6: Optimize the overall objective in (10) through stochastic gradient descent.
7: Update memory bank features and output probabilities.
end while

space:

$$\hat{q}_j^m = \frac{1}{K} \sum_{\ell \in \mathcal{N}_j} p_\ell, \tag{6}$$

where $\mathcal{N}_j$ is the index of the nearest neighbors for the j-th sample. m denotes different feature spaces, i.e., image, text and multimodal spaces.

Then, we can directly use the maximum probability prediction corresponding to this probability as the structural pseudo label for each space:

$$\hat{y}_j^m = \arg\max_k \hat{q}_{j,k}^m. \tag{7}$$

After obtaining the semantic pseudo label $\hat{y}_j^{se}$ and structural pseudo labels $\hat{y}_j^m$, we implement the multi-view consensus mechanism through a consistency checking process. We consider the pseudo label reliable when all views' pseudo labels from available modalities are consistent. This is crucial for our adaptive modality mechanism—when a certain modality is missing, we exclude it from the consistency checking process, as including artificially generated features may introduce noise and potentially decrease reliability. This approach adapts dynamically to different scenarios, enabling cross-platform adaptability without requiring all modalities to be present. The consistency loss can be represented as:

$$\ell_{co} = -\frac{1}{n_{co}} \sum_{j=1}^{n_{co}} \log p_{j,\hat{y}_j}, \tag{8}$$

where $\hat{y}_j$ is the reliable pseudo label from voting for the j-th sample. n_{co} denotes the number of samples with a consistent pseudo label.

However, although the quality of pseudo labels obtained through consistency samples is high, even in the absence of modalities, the number of samples available for training has decreased after screening. Observations from preliminary experiments revealed a noteworthy phenomenon: after several rounds of training, the ground-truth label of an inconsistent sample is likely to be found among these pseudo labels generated from different perspectives. This observation led us to develop a self-penalization mechanism that leverages partial information from inconsistent samples. Rather than requiring perfect agreement, we utilize the collective "negative knowledge" from these samples—when multiple views disagree on the exact label but collectively rule out certain classes, this exclusion pattern provides valuable negative supervision. By penalizing categories not present in any of the generated pseudo-labels, we could increase the likelihood of the model predicting the correct category while making use of all available training data. The self-penalization loss can be formulated as:

$$\ell_{sp} = \frac{1}{n_t - n_{co}} \sum_{i=j}^{n_t - n_{co}} \log(1 - p_{j,\mathbb{1}_j}), \tag{9}$$

where $\mathbb{1}_j$ is a one-hot label, with 0 at the j-th position when it is in the extracted pseudo label list and 1 elsewhere.

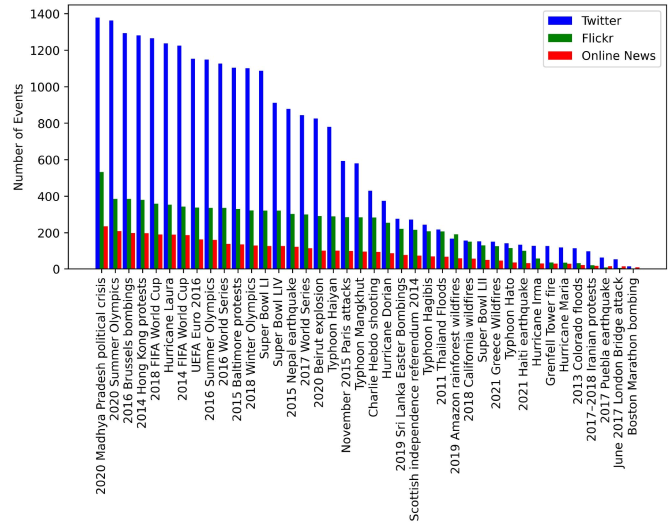


Fig. 5. The distribution of CNED dataset.

E. Overall Objective

The overall objective function to be minimized can be formulated as follows:

$$\ell = \ell_c + \alpha\ell_{mc} + \beta\ell_d + \lambda(\ell_{co} + \ell_{sp}), \tag{10}$$

$$\ell_c = -\frac{1}{n_s} \sum_{i=j}^{n_s} \log p_{j,\hat{y}_j^s}, \tag{11}$$

where ℓ_c denotes the loss for the source domain. α and β are hyperparameters. To reduce parameter sensitivity and ease the selection of models like [41], we adopt a gradual progressive strategy for λ. λ is set to $\frac{2}{1+\exp(-10 \cdot p)} - 1$ where p is the training progress linearly changing from 0 to 1. This gradual increase attenuates the influence of noise inherent in the pseudo labels during the preliminary iterations, consequently preventing the accumulation of errors to some degree.

The detailed algorithm for the SSMC method is presented in Algorithm 1.

V. Experiment

A. Cross-Platform News Event Dataset (CNED)

1) Collection of News Events: Our dataset collects significant news events from the past decade from Wikipedia, as Wikipedia, being a crowd-sourced platform, ensures that all news event entries are manually verified. Specifically, we select these events based on a variety of event themes (for example, natural disasters, sports events, political elections, etc.), events with influence (widely spread across major platforms) and events of similar topics (for example, typhoons in different regions, sports events at different times, etc.). Eventually, we collected 40 news event entries, as shown in Fig. 5.

2) Collection and Statistics of the Dataset: We chose three mainstream social media platforms for data collection, including the multimodal Twitter platform (T), the image-focused Flickr platform (F), and various online news platforms (O) that

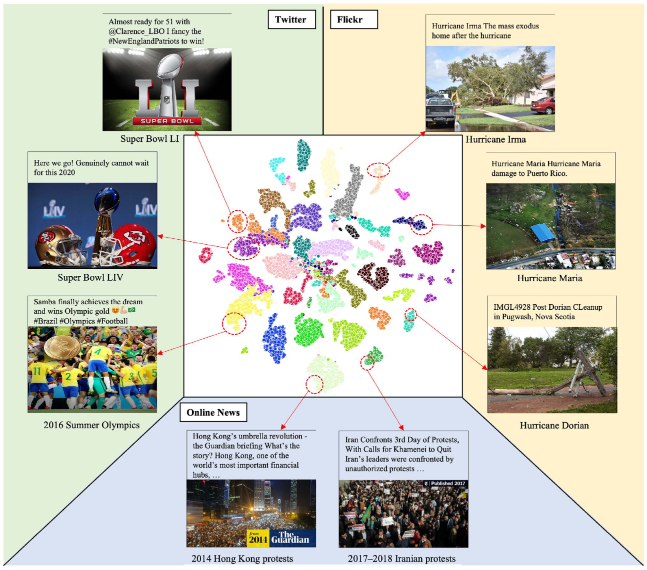


Fig. 6. Feature visualization of the CNED dataset for different platforms.

primarily feature long texts. Specifically, to avoid task simplification through direct keyword search, for Twitter, we use event-related hashtags and the time of the news event for collection; for Flickr, we utilize event-related keywords to search for related album sets, then collect related posts from them; for online news, we collect through the related links in the corresponding Wikipedia entries. Ultimately, we filtered all single-modal samples and manually checked whether the data semantically matches the corresponding events, resulting in 37,711 samples. Specifically, unlike previous datasets, we did not filter out non-English samples, considering that multilingual data could provide a more comprehensive perspective of news events. The statistics, distribution and feature visualization of our collected CNED dataset are shown in Table I, Figs. 5 and 6, respectively.

3) Comparisons With Existing Datasets: Most of the current news event datasets are based on single-platform data, including both single-modal datasets [46], [47], [48] and multimodal datasets [9], [49], [50], [51]. Moreover, the datasets they collect are mostly in English only, which, for the task of news event detection, lacks the interpretation of different perspectives on events. Additionally, most datasets are directly collected based on keywords, which diminishes the value of multimodal data because texts retrieved solely by keywords can already perform well. These datasets are difficult to use for cross-platform news event detection directly as the news events are different. Therefore, we have collected a dataset that is multi-platform, multimodal, and multilingual, hoping to advance the development of this field.

B. Baselines

We compare SSMC across two distinct scenarios: complete multimodal cross-platform news event detection (with a missing rate of 0%) and cross-platform news event detection with missing modalities (missing rates of 20% and 40%, respectively).

For the first scenario, our baselines include various domain adaptation methods as our benchmarks. Specifically, CORAL [29] aligns the second-order statistics of the source and target distributions through a linear transformation. DAN [30] uses a multiple kernel variant of maximum mean discrepancies to learn transferable features in deep networks. JAN [31] aims to learn a transfer network by aligning the joint distributions of multiple domain-specific layers across domains using a joint maximum mean discrepancy criterion. DANN [34] focuses on learning domain-invariant features by adversarial learning. MMDA [33] employs multiple adversarial losses to learn common multimodal features. MCC [37] introduces a minimum class confusion constraint to achieve transfer learning. ATDOC [36] alleviates classifier bias by introducing an auxiliary classifier specifically for target data, thereby improving the quality of pseudo labels. ENT [38] integrates domain adversarial training into entropy minimization to enhance pseudo-label accuracy. CDCL [42] presents a method based on contrastive self-supervised learning aimed at feature alignment to minimize the domain gap between the training and testing datasets. RCE [43] introduces a training method that aligns with risk consistency, allowing the model to learn information from noisy pseudo-labeled data without compromising the performance. Additionally, we report the performance of direct training on the source platform without applying domain adaptation methods, i.e., image-only, text-only, and image+text configurations. We also report the results of training and testing using the target domain, which serves as the upper bound (following an 8:2 split between the training and testing sets on the target platform). Since some baselines are primarily applied in the image domain, we use the same CLIP for extracting image and text features in our experiments. These features are then concatenated to form the multimodal features for adaptation.

For the second scenario, our baselines involve methods tailored for missing modalities. In detail, DAL [44] proposes incorporating available category information and adversarial training to enable the model to generate more informative domain information despite missing modalities. DVAE [45] introduces dual-aligned variational autoencoders to learn modality-invariant representations, addressing the challenge of missing modalities in data. AT-CVAE [17] proposes an adaptive transformer-based conditioned variational autoencoder network for incomplete social event classification, leveraging the capabilities of variational autoencoders and transformers to handle incomplete data.

C. Implementation Details

We evaluate all cross-platform scenarios (i.e., $T \rightarrow F$, $F \rightarrow T$, $T \rightarrow O$, $O \rightarrow T$, $F \rightarrow O$ and $O \rightarrow F$). For scenarios with missing modalities, we randomly mask the images or the text with a missing rate on both source and target domains. The missing rate is set at 0%, 20%, and 40%. Take 20% as an example (20% of the samples only contain text, 20% of the samples only contain images, and 60% of the samples contain both images and text for both source and target platforms).

Our implementation uses pretrained CLIP encoders [39] with ViT-B-32 that produce 512-dimensional feature vectors as initial

TABLE II
ACCURACY (ACC) AND F1 SCORE ON CNED DATASET FOR CROSS-PLATFORM MULTIMODAL NEWS EVENT DETECTION

Methods	$T \to F$		$F \to T$		$T \to O$		$O \to T$		$F \to O$		$O \to F$		Average	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Missing Rate: 0%														
Image Only	0.470	0.413	0.481	0.385	0.441	0.411	0.485	0.426	0.418	0.359	0.467	0.418	0.460	0.402
Text Only	0.669	0.658	0.593	0.532	0.784	0.764	0.653	0.602	0.828	0.793	0.725	0.712	0.709	0.677
Image+Text	0.761	0.685	0.669	0.595	0.784	0.753	0.692	0.645	0.818	0.767	0.804	0.748	0.755	0.699
CORAL [29]	0.766	0.682	0.667	0.596	0.780	0.749	0.689	0.642	0.813	0.766	0.805	0.750	0.753	0.697
DAN [30]	0.749	0.663	0.675	0.582	0.780	0.749	0.690	0.630	0.785	0.735	0.807	0.738	0.748	0.683
JAN [31]	0.724	0.645	0.647	0.539	0.694	0.679	0.637	0.606	0.723	0.676	0.788	0.713	0.702	0.643
DANN [34]	0.737	0.660	0.667	0.587	0.776	0.749	0.684	0.638	0.803	0.672	0.803	0.735	0.745	0.674
MMDA [33]	0.741	0.667	0.684	0.592	0.781	0.748	0.688	0.634	0.783	0.730	0.807	0.741	0.747	0.685
MCC [37]	0.763	0.676	0.665	0.577	0.811	0.762	0.709	0.625	0.825	0.774	0.792	0.714	0.761	0.688
ATDOC [36]	0.724	0.666	0.718	0.640	0.775	0.743	0.686	0.658	0.794	0.749	0.745	0.684	0.740	0.690
ENT [38]	0.743	0.675	0.668	0.628	0.791	0.763	0.713	0.655	0.749	0.748	0.800	0.727	0.744	0.699
CDCL [42]	0.810	0.741	0.713	0.635	0.855	0.813	0.716	0.649	0.824	0.781	0.802	0.744	0.787	0.727
RCE [43]	0.826	0.742	0.704	0.620	0.843	0.795	0.693	0.630	0.812	0.766	0.806	0.727	0.781	0.713
SSMC	**0.839**	**0.762**	**0.746**	**0.670**	**0.857**	**0.819**	**0.749**	**0.680**	**0.826**	**0.797**	**0.814**	**0.769**	**0.805**	**0.750**
Upper Bound	0.957	0.897	0.893	0.841	0.934	0.921	0.893	0.841	0.934	0.921	0.957	0.897	0.928	0.886
Missing Rate: 20%														
DAL [44]	0.648	0.573	0.580	0.493	0.682	0.643	0.627	0.575	0.639	0.535	0.695	0.618	0.645	0.573
DVAE [45]	0.665	0.594	0.594	0.510	0.702	0.656	0.608	0.553	0.698	0.632	0.690	0.621	0.659	0.594
AT-CVAE [17]	0.683	0.612	0.609	0.525	0.706	0.676	0.625	0.577	0.716	0.660	0.706	0.655	0.674	0.617
SSMC	**0.736**	**0.674**	**0.686**	**0.601**	**0.779**	**0.734**	**0.681**	**0.607**	**0.737**	**0.693**	**0.740**	**0.686**	**0.727**	**0.666**
Missing Rate: 40%														
DAL [44]	0.506	0.554	0.519	0.436	0.609	0.576	0.557	0.503	0.575	0.476	0.595	0.542	0.561	0.514
DVAE [45]	0.555	0.506	0.518	0.436	0.618	0.585	0.536	0.480	0.608	0.566	0.589	0.537	0.571	0.518
AT-CVAE [17]	0.609	0.563	0.542	0.454	0.641	0.614	0.558	0.505	0.634	0.582	0.616	0.575	0.600	0.549
SSMC	**0.644**	**0.603**	**0.611**	**0.530**	**0.692**	**0.645**	**0.597**	**0.535**	**0.653**	**0.602**	**0.652**	**0.612**	**0.641**	**0.588**

TABLE III
ABLATION STUDY

#	ℓ_{mc}	ℓ_d	ℓ_{sp}	ℓ_{co}	M	G	Accuracy	F1
1	×	✓	✓	✓	×	✓	0.717	0.636
2	✓	×	✓	✓	×	✓	0.723	0.639
3	×	×	✓	✓	×	✓	0.709	0.630
4	✓	✓	×	✓	×	✓	0.722	0.652
5	✓	✓	✓	×	×	✓	0.715	0.650
6	✓	✓	×	×	×	✓	0.708	0.649
7	✓	✓	✓	✓	×	×	0.693	0.622
8	✓	✓	✓	✓	✓	×	0.709	0.636
9	✓	✓	✓	✓	×	✓	**0.736**	**0.674**

M and G indicate M2M-100 model [52] and Google API respectively ($T \to F$, missing rate: 20%).

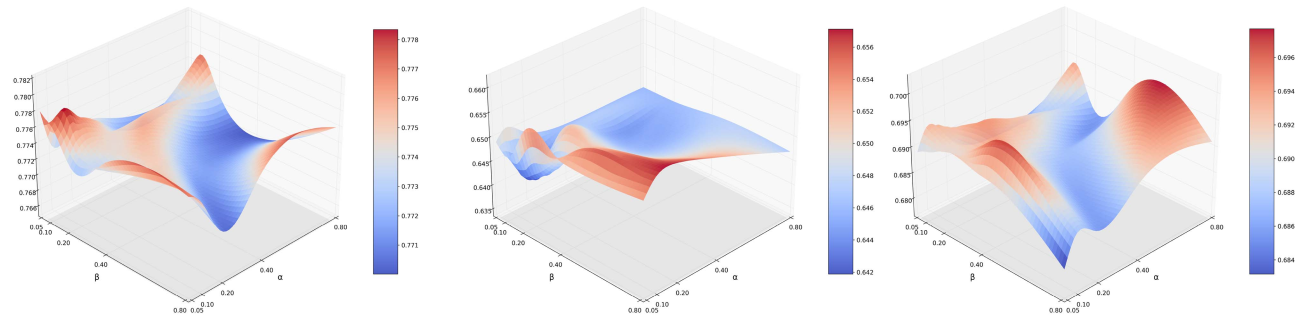
representations. The modality-specific layers are implemented as two-layer MLPs with dimensions (512$\to$ 256 $\to$256). The modality-shared layers follow the same structure with identical dimensions. The projection layer is constructed as an MLP with dimensions (512$\to$ 256 $\to$256). The modality classifier consists of a two-layer MLP with dimensions (256$\to$ 256 $\to$2), with a sigmoid activation at the output to predict the modality type. The final classification layer is an MLP with dimensions (512$\to$ 256 $\to C$), where C represents the number of events in our task. Most MLPs in our architecture employ ReLU activation functions after each layer (except output layers) and include dropout regularization with a probability of 0.5, while output layers use task-appropriate activations (i.e. sigmoid for modality classification, softmax for final classification).

For O and F, we combine the title and content as the text content. For MDC, α and β are set to 0.1 based on the observation of the performance of the labeled source domain as the target domain is unlabeled. In fact, they can be arbitrarily selected within a certain order of magnitude; our subsequent parameter analysis demonstrates that the results are robust to some variations of these hyperparameters. For MSL, we set the sharpening parameter t to 5 for memory bank construction, and the number of nearest neighbors N to 3 for pseudo label generation. We employ Adam as the optimizer, the learning rate as 1e-4, the batch size as 40, and the training epochs as 30. Accuracy and F1 score are used as the evaluation metric. In the experiment, we randomly choose five different seeds and compute their average to obtain the final result. All baselines use the Google Translation API for fairness in comparison.

D. Comparison With the State of the Arts

Table II summarizes the performances of various methods on the CNED dataset. From the results, we have the following observations:

- Compared with other domain adaptation and missing modality methods, our method can simultaneously handle cross-platform and incomplete modalities and achieve the best performance, indicating our model's strong generalization ability for the cross-platform multimodal news event detection task.
- Many domain adaptation methods have declined results compared to those not using domain adaptation, possibly due to a large domain gap for the cross-platform multimodal news event detection task causing negative transfer.
- As the missing rate increases, the degree of decline in our method is relatively small compared to other models, which demonstrates the effectiveness of the proposed MDC.
- Our model performs relatively poorly when the target platform is Twitter compared to other platforms. This is due to Twitter having a larger number of posts and a wider variety



(a) Twitter (b) Twitter→Flickr (c) Twitter→Online News

Fig. 7. 3D visualization of accuracy in parameter sensitivity analysis across different scenarios. (a) shows the performance on the source domain (Twitter), while (b) and (c) show the performance on the target domains (Twitter → Flickr and Twitter → Online News, respectively) under a missing modality rate of 40%.

of languages compared to other social media platforms, which inevitably contains more noise.

E. Ablation Study

To investigate the effectiveness of the various modules, we conduct an ablation experiment. Specifically, we consider removing each of our proposed modules to test the performance, i.e., without (w/o) the modal classification loss ℓ_{mc}, confusion loss ℓ_d, self-penalization loss ℓ_{sp} and consistency loss ℓ_{co}. As illustrated in Table III, we select one of the cross-platform scenarios with missing rate as 20%, i.e., $T\rightarrow F$. From the table, we observe that:

- The performance of our model declines after removing any module, demonstrating the effectiveness of each module we proposed.
- The significant impact is on the consistency loss, as it provides high-quality pseudo labels that directly facilitate learning in the target domain.

Furthermore, we also conducted ablation studies on the translation module. Although our paper does not focus on this part, it exists in our dataset, and here we show the impact of this part on our model's results. Specifically, we choose another multilingual translation model, i.e., M2M-100 [52], for comparison. As shown in Table III, we observe that the Google API performs better than the M2M-100 model, which is why we select it as our preprocessing module.

F. Impact of Parameters α and β

We analyze the impact of parameters α and β on model performance by expanding the parameter range to $[0.05, 0.8]$ across three scenarios: source domain (Twitter) and two cross-platform settings (Twitter $\rightarrow$ Flickr and Twitter $\rightarrow$ Online News) with a 40% missing modality rate. As shown in Fig. 7, the optimal parameters for the source domain are $\alpha = \beta = 0.1$ (accuracy 0.778). Interestingly, while the target domains have slightly different optimal values (Twitter $\rightarrow$ Flickr: $\alpha = 0.2$, $\beta = 0.8$ with accuracy 0.657; Twitter $\rightarrow$ Online News: $\alpha = 0.1$, $\beta = 0.4$ with accuracy 0.696), using the source-optimized parameters ($\alpha = \beta = 0.1$) still achieves good performance on target domains (0.644 and 0.692 respectively, with differences of only

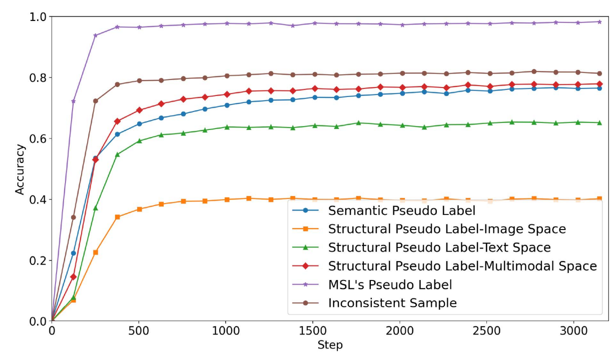


Fig. 8. Accuracy of pseudo label during training ($T\rightarrow F$, missing rate: 20%).

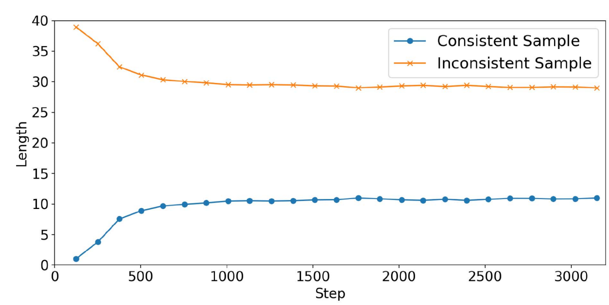
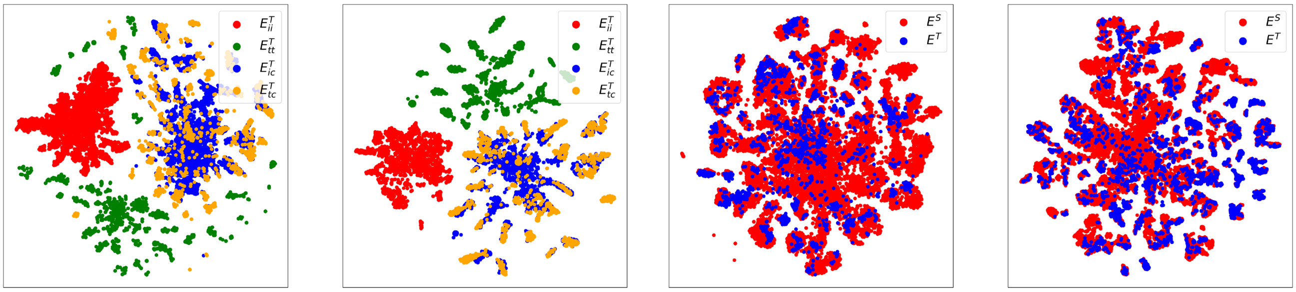


Fig. 9. Length of consistent and inconsistent samples during training ($T\rightarrow F$, missing rate: 20%, batch size: 40).

1.3% and 0.4%). This demonstrates that parameters selected using only source domain data (as is necessary in real-world scenarios where target labels are unavailable) can effectively generalize to target domains.

G. Evaluating the Effectiveness of MSL

To further explore the effectiveness of the consistency loss and self-penalization loss proposed in our MSL, we observe the accuracy of pseudo labels from different perspectives on the target domain during the training process. As shown in Fig. 8, we find that compared to other pseudo labels, our method's consistency pseudo labels achieve an accuracy rate close to 100% after several rounds of training, even with a missing



(a) $T \to F$ (b) $T \to O$ (c) AT-CVAE (d) SSMC

Fig. 10. Visualization of the feature distribution (missing rate: 20%). (a) and (b): Common features and modality-specific features from the target domain under $T \to F$ and $T \to O$; (c) and (d): Multimodal features from source and target domains under $T \to O$.

(a) (b) (c) (d)

Fig. 11. Success and failure examples induced by SSMC on the CNED dataset from two different scenarios, i.e., $T \to F$ and $T \to O$ (missing rate: 20%).

rate of 20%, which proves the high quality of our proposed pseudo labels. Meanwhile, as shown in Fig. 9, the proportion of these high-quality samples is only 25% as mentioned in Section IV-D. Therefore, we use the self-penalization loss to learn from the remaining inconsistent samples. In Fig. 8, we also visualized the accuracy between inconsistent samples' labels and the pseudo labels list obtained from different perspectives. We can find the accuracy rate is relatively high (around 80%) compared to other pseudo labels, which verifies our previous assumption. Therefore, by combining these two strategies, our model can utilize all samples for training, which results in good performance.

H. Visualization

To validate the effectiveness of MDC, we employ t-SNE to visualize the common features and the modality-specific features from the target domain. As shown in Figs. 10(a) and (b), we can observe that the common features of images and texts (blue and yellow) are thoroughly mixed together, demonstrating successful modality-agnostic representation in the shared space. In contrast, the modality-specific features (red and green) exhibit clear separation with minimal overlap between modalities, confirming that our adversarial training mechanism effectively establishes an information boundary. This visualization provides empirical evidence that our MDC module successfully prevents information leakage, as the distinct clustering patterns in modality-specific features and the intermixed patterns in shared features directly reflect the constraints imposed by our complementary loss functions (ℓ_{mc} and ℓ_d).

In addition, we also visualize the multimodal features from both the source and target domains by using different methods. As illustrated in Fig. 10(c) and (d), the boundaries of our SSMC's features from the source and target domains are clearer compared to AT-CVAE, which validates the effectiveness of our proposed MSL module.

I. Case Study

Fig. 11 presents several success and failure examples in different scenarios. From the figure, we have the following observations:

- Fig. 11(a) and (c) illustrate successful predictions by SSMC, which demonstrates that our model can make correct judgments based on another modality when one modality is missing.
- As shown in Fig. 11(b), when the target platform lacks text, if the image information does not contain more elements about the corresponding news event, our model is prone to errors (e.g., although it predicts a typhoon event based on the collapse of trees, it is still a different event).
- As illustrated in Fig. 11(d), When the target platform lacks images, the lengthy texts in online news may contain many descriptions unrelated to the event, which misleads our model (e.g., the example describes extensively the causes of a fire without describing wildfires).

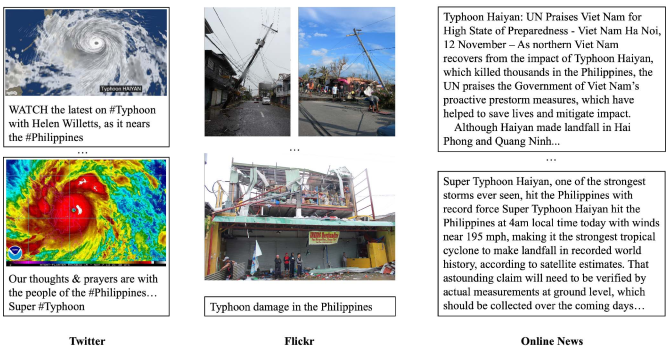


Fig. 12. Samples detected by using SSMC for the "Typhoon Haiyan" event with Twitter as the source platform and Flickr and Online News as the target platforms.

In addition, to validate the importance of cross-platform event detection (including improving the quality of event data from a single platform), we visualize some examples using the SSMC model. Specifically, we take Twitter as the source platform, and Flickr and Online News as the target platform to detect the "Typhoon Haiyan" event. As shown in Fig. 12, we can find that Twitter, as the source platform, mainly consists of real-time updates and alert information rapidly posted by users during the natural disaster. The detected samples from target platforms like Flickr focus more on high-quality post-disaster photos showing the damage, while online news provides textual descriptions of the entire natural disaster. Therefore, by combining data from cross-platform sources, we can overcome the limitations of a single platform and obtain more comprehensive information, which validates the importance of cross-platform event detection.

VI. Conclusion

In this paper, we have proposed a Self-Supervised Modality Complementation (SSMC) method that effectively addresses the challenges of missing modality and cross-platform scenarios in the cross-platform multimodal news event detection task. A missing data complementation module is designed to use modality-shared features to supplement missing modality scenarios, and a multimodal self-learning module generates reliable pseudo labels from multiple perspectives to achieve self-learning and self-penalization in the target domain. We have also introduced a comprehensive Cross-platform News Event Detection (CNED) dataset, encompassing diverse platforms and a wide range of public news events. Experimental results on the CNED dataset validate the effectiveness of our proposed method, demonstrating its competitive performance in the cross-platform multimodal news event detection task compared to state-of-the-art methods.

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